

OLIST E-COMMERCE DATA ANALYSIS

Final Project Report • Python • Pandas • Matplotlib • Seaborn • PostgreSQL • SQL • Jupyter

1. Executive Summary

This project presents an end-to-end e-commerce data analytics workflow using the Olist dataset. The project starts with raw CSV data, performs data loading, auditing and exploratory analysis in Python/Jupyter, uses Pandas for data manipulation, Matplotlib and Seaborn for analytical visualization, and PostgreSQL for structured business analysis through SQL. The goal is to convert transactional data into clear business questions, measurable KPIs and actionable operational insights.

2. Project Objectives

- Audit and understand the structure and quality of the e-commerce datasets.
- Analyze customer, order, product, payment and delivery behavior.
- Use statistical aggregation and visualization to identify important patterns.
- Build a PostgreSQL database and perform business-oriented SQL analysis.
- Document the complete workflow in a reproducible Jupyter Notebook and SQL file.
- Prepare a portfolio-ready project suitable for GitHub and a Data Analyst resume.

3. Technology Stack

Technology	Role in the Project
Python	Core programming and analysis workflow
Pandas	Data loading, cleaning, transformation and aggregation
Matplotlib	Data visualization and trend analysis
Seaborn	Statistical and categorical visualizations
Jupyter Notebook	Interactive analysis and documentation
PostgreSQL	Relational database and analytical storage
SQL / pgAdmin	Business queries, aggregation and relational analysis
Git / GitHub	Portfolio version control and project sharing

4. Dataset & Database Structure

The project uses the major Olist analytical tables. The datasets were loaded into PostgreSQL after being handled through Python/Jupyter. The following row counts were verified during the database loading process.

Table	Rows
orders	99,441
order_items	112,650
products	32,951
customers	99,441
order_payments	103,886

5. Python / Pandas Data Analysis

Python and Pandas were used as the primary data-analysis layer. The notebook loads the source data, checks the structure and performs exploratory calculations before the data is used for database analysis.

- Loaded multiple Olist CSV datasets using Pandas.
- Inspected row counts, columns and data types.
- Performed aggregation and category-level analysis.
- Calculated delivery-time and late-delivery metrics.
- Prepared analytical outputs for interpretation and reporting.

6. Visualization: Matplotlib & Seaborn

Matplotlib and Seaborn were used to turn analytical results into readable visual evidence. These libraries are important in the project because the objective is not only to calculate metrics, but also to identify patterns and communicate them clearly.

- Matplotlib — used for plotting analytical trends and category-level comparisons.
- Seaborn — used for cleaner statistical and categorical visualizations.
- Visual analysis — used to compare delivery performance and identify categories requiring attention.

7. Key Python Analysis Findings

The completed notebook produced the following category-level delivery findings. These values are retained as project evidence and should be referenced in the final GitHub documentation.

Finding	Observed Result
Highest average delivery time	moveis_escritorio — 20.84 days
2nd highest average delivery time	artigos_de_natal — 15.74 days
3rd highest average delivery time	fashion_calcados — 15.44 days
Highest late-delivery rate	casa_conforto_2 — 16.67%
2nd highest late-delivery rate	moveis_colchao_e_estofado — 13.51%
3rd highest late-delivery rate	audio — 12.71%

8. Business Interpretation

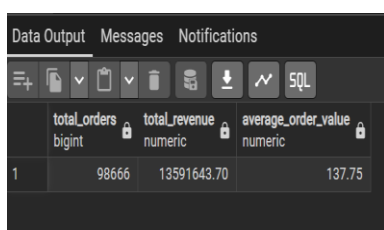
Delivery performance varies materially across product categories. Categories with higher average delivery times and higher late-delivery rates may require further investigation of fulfillment processes, logistics partners, product handling, inventory availability or delivery expectations. Category-level monitoring can therefore provide more actionable insight than relying only on an overall average.

9. PostgreSQL & SQL Business Analysis

The project translates business requirements into SQL questions. The following questions are the final business-analysis checklist used in the SQL stage. The SQL outputs can be pasted below each question in the project documentation or GitHub README.

SQL Question 1

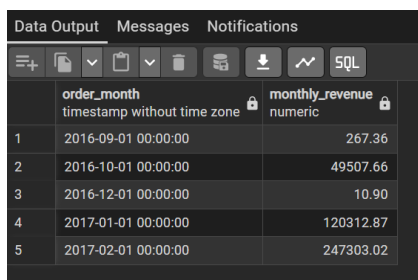
Business Question: What is the total number of orders, total revenue, and average order value?



	total_orders bigint	total_revenue numeric	average_order_value numeric
1	98666	13591643.70	137.75

SQL Question 2

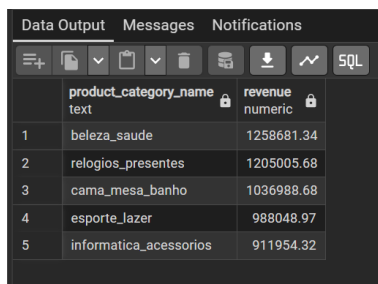
Business Question: How does monthly revenue change over time?



	order_month timestamp without time zone	monthly_revenue numeric
1	2016-09-01 00:00:00	267.36
2	2016-10-01 00:00:00	49507.66
3	2016-12-01 00:00:00	10.90
4	2017-01-01 00:00:00	120312.87
5	2017-02-01 00:00:00	247303.02

SQL Question 3

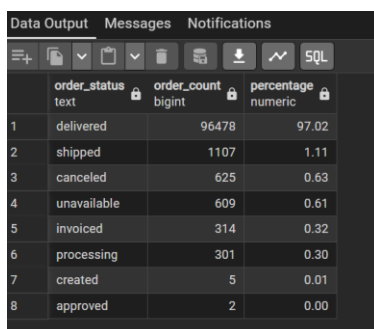
Business Question: Which product categories generate the highest revenue?



	product_category_name text	revenue numeric
1	beleza_saude	1258681.34
2	relogios_presentes	1205005.68
3	cama_mesa_banho	1036988.68
4	esporte_lazer	988048.97
5	informatica_acessorios	911954.32

SQL Question 4

Business Question: What is the distribution of orders across different order statuses?



	order_status text	order_count bigint	percentage numeric
1	delivered	96478	97.02
2	shipped	1107	1.11
3	canceled	625	0.63
4	unavailable	609	0.61
5	invoiced	314	0.32
6	processing	301	0.30
7	created	5	0.01
8	approved	2	0.00

SQL Question 5

Business Question: What percentage of customers are repeat customers?

- JOIN operations across related business tables.
- GROUP BY and ORDER BY for category and time-based analysis.
- Date/time functions for monthly and delivery analysis.
- Conditional aggregation and percentage calculations.
- CTEs for customer-level repeat-order analysis.
- Window functions for percentage and distribution calculations.

11. End-to-End Workflow

Stage	Completed Work
1. Data Loading	Loaded Olist CSV datasets with Python/Pandas
2. Data Audit	Inspected datasets and validated analytical structure
3. EDA	Calculated business metrics and category-level patterns
4. Visualization	Used Matplotlib and Seaborn for analytical visuals
5. Database	Loaded core tables into PostgreSQL
6. SQL Analysis	Created 8 business-focused SQL questions/queries
7. Documentation	Prepared project report and GitHub-ready structure

12. Business Recommendations

- Investigate categories with the highest average delivery time.
- Prioritize categories with elevated late-delivery rates for logistics review.
- Track delivery KPIs at category level to identify operational bottlenecks.
- Combine revenue, customer, payment and delivery analysis when making business decisions.
- Use the SQL outputs and visual analysis together rather than relying on a single metric.

13. GitHub Deliverables

- `01\_data\_audit.ipynb` — Python/Pandas data audit and analysis notebook.
- `olist\_business\_analysis.sql` — PostgreSQL business-analysis queries.
- `README.md` — project overview, methodology, findings and business context.
- `Olist\_Ecommerce\_Final\_Project\_Report.pdf` — complete project documentation.

14. Resume-Ready Project Entry

Olist E-commerce Sales & Delivery Analysis | Python, Pandas, Matplotlib, Seaborn, SQL, PostgreSQL

Analyzed 99K+ e-commerce orders and related customer, product, payment and order-item data using Python/Pandas; performed exploratory analysis with Matplotlib and Seaborn, loaded analytical tables into PostgreSQL, and developed 8 business-focused SQL analyses covering revenue, customer behavior, product categories, payments and delivery performance.

15. Current Project Status

Completed: Python/Pandas analysis, Matplotlib/Seaborn visualization, PostgreSQL setup, data loading, SQL business analysis, project documentation and GitHub preparation.

